COMP4358 - Cloud Computing

Lab 6: Load Balancing with Application Gateways and Traffic Filter

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In this lab, I set up geographic load balancing on Microsoft Azure using Application Gateways and Traffic Manager. I deployed virtual machines in two regions (US and France), configured regional Application Gateways to distribute traffic across the VMs, and used Traffic Manager with Geographic routing so that users are automatically directed to the correct regional gateway based on their location.

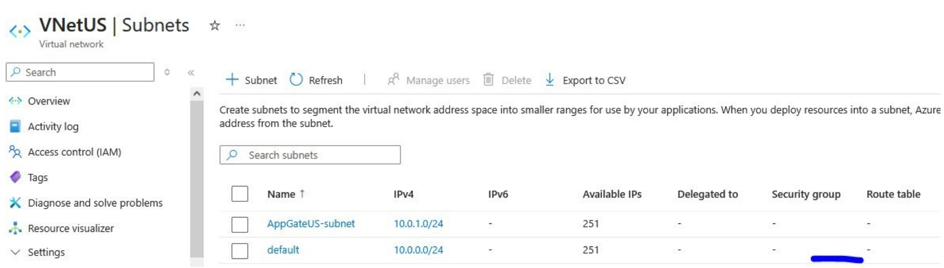
Screenshots

Resource Groups

Two resource groups were created: LabUS-RG in West US 2 for all US side resources, and LabFrance-RG in West Europe for all France side resources.

VNetUS Subnets

VNetUS was created with address space 10.0.0.0/16. Two subnets were configured: AppGateUS-subnet (10.0.1.0/24) for the Application Gateway, and default (10.0.0.0/24) for the VMs.

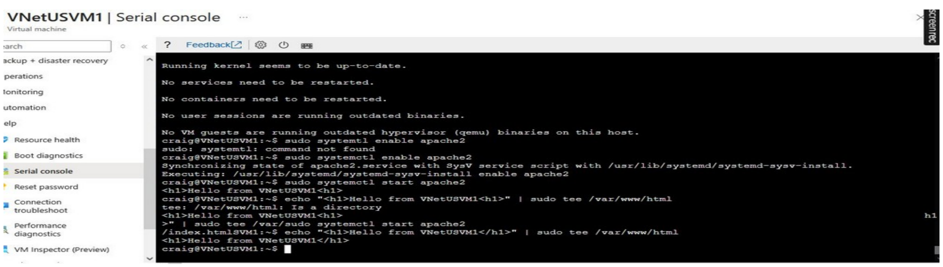


US Virtual Machines

VNetUSVM1 and VNetUSVM2 were deployed in LabUS-RG in West US 2 with no public IPs. Both VMs are in the default subnet of VNetUS and are showing Running status.

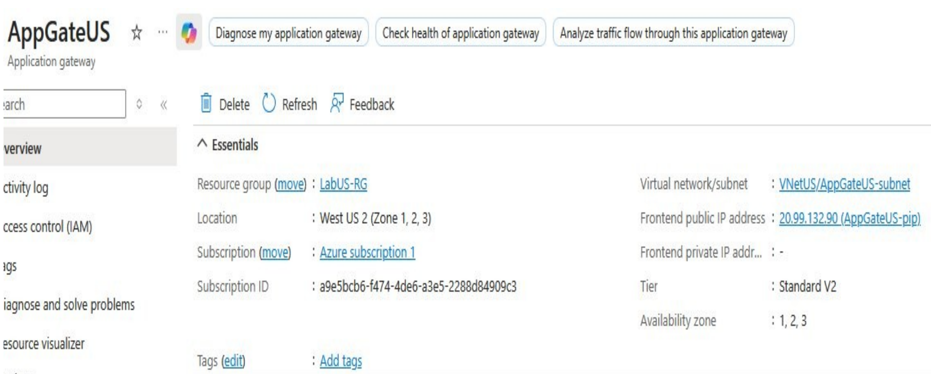
Apache Installation via Serial Console

Apache was installed on VNetUSVM1 using the Serial Console. The web server was enabled, started, and a custom HTML page was written to identify the VM in browser responses.



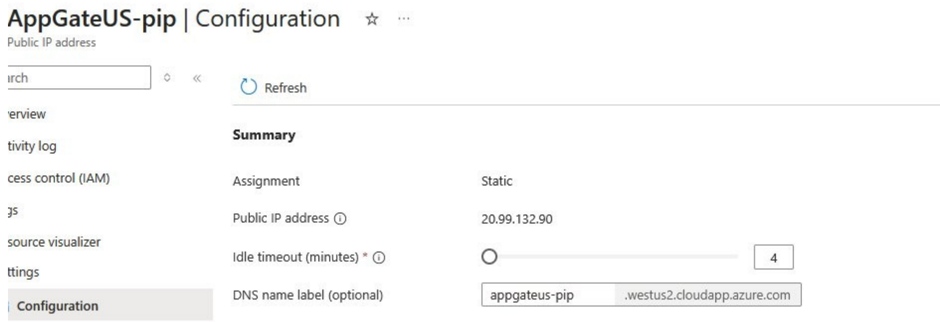
AppGateUS Overview

AppGateUS was deployed in West US 2 using the AppGateUS-subnet of VNetUS. The frontend public IP is 20.99.132.90 (AppGateUS-pip) and the tier is Standard V2.



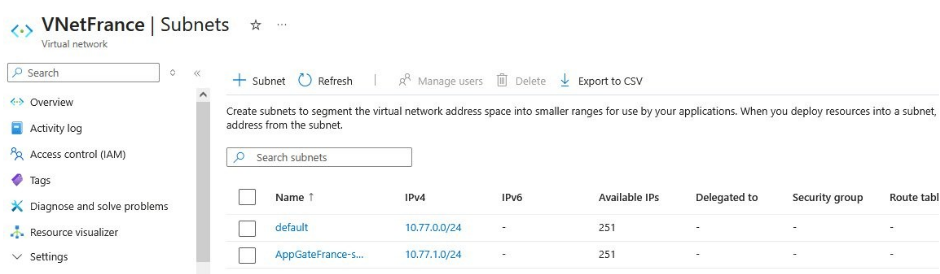
AppGateUS Public IP DNS Name Label

A DNS name label was added to the AppGateUS-pip public IP resource. The label appgateus-pip was assigned, producing the hostname appgateus-pip.westus2.cloudapp.azure.com. This step is required for Traffic Manager to accept the IP as an endpoint.



VNetFrance Subnets

VNetFrance was created with address space 10.77.0.0/16 in West Europe. Two subnets were configured: AppGateFrance-subnet (10.77.1.0/24) for the Application Gateway, and default (10.77.0.0/24) for the VMs.

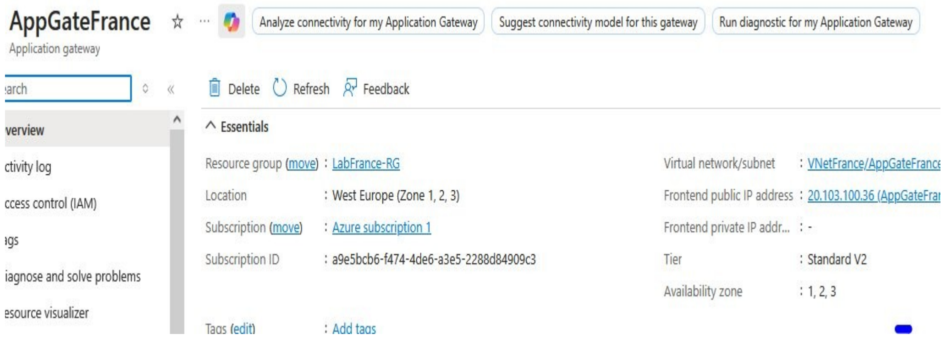


France Virtual Machines

VNetFranceVM1 and VNetFranceVM2 were deployed in LabFrance-RG in West Europe with no public IPs. Both VMs are in the default subnet of VNetFrance and are showing Running status.

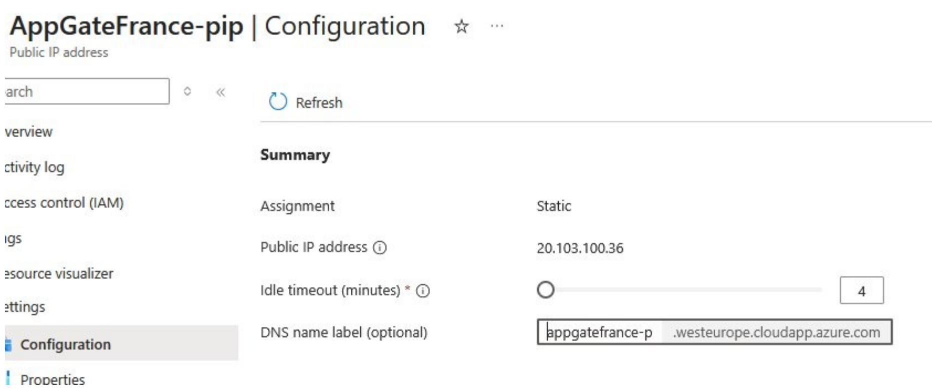
AppGateFrance Overview

AppGateFrance was deployed in West Europe using the AppGateFrance-subnet of VNetFrance. The frontend public IP is 20.103.100.36 (AppGateFrance-pip) and the tier is Standard V2.



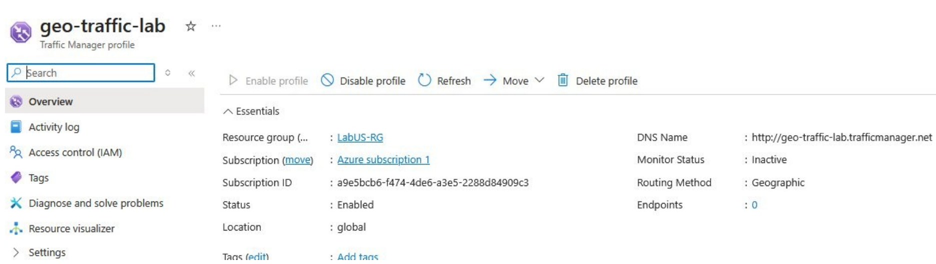
AppGateFrance Public IP DNS Name Label

A DNS name label was added to the AppGateFrance-pip public IP resource. The label appgatefrance-p was assigned, producing the hostname appgatefrance-p.westeurope.cloudapp.azure.com.



Traffic Manager Profile

A Traffic Manager profile named geo-traffic-lab was created with Geographic routing method. The DNS name is http://geo-traffic-lab.trafficmanager.net and the profile status is Enabled.



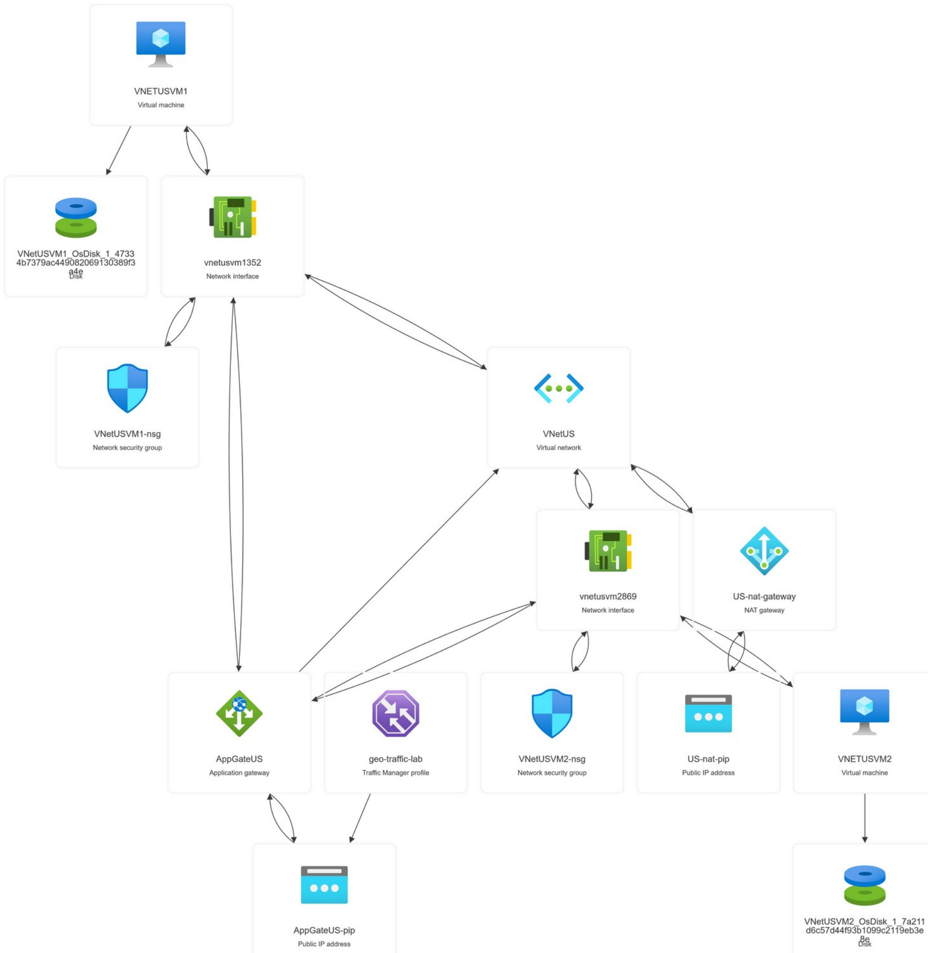
Traffic Manager Endpoints

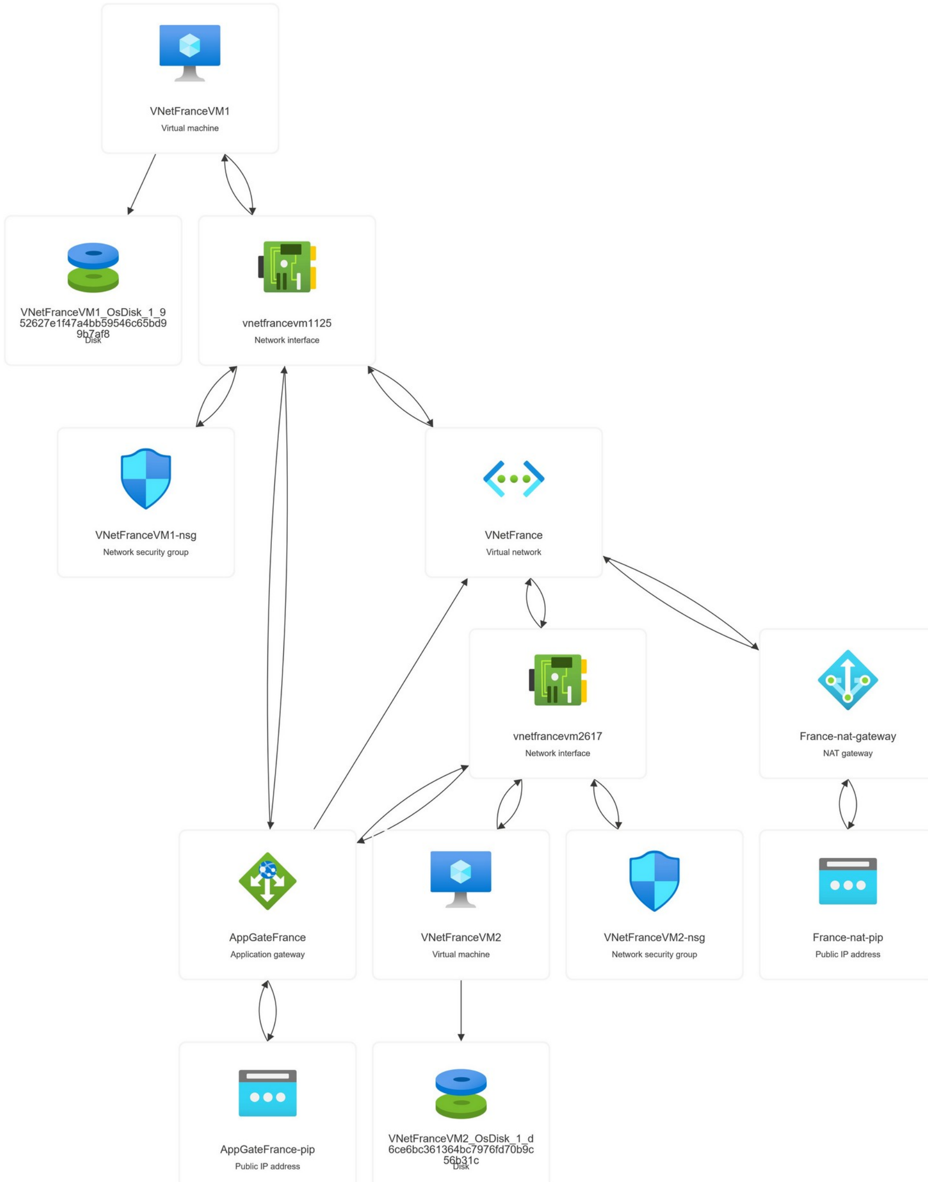
Two endpoints were added to the Traffic Manager profile. US-endpoint is mapped to the United States geographic region pointing to AppGateUS-pip. France-endpoint is mapped to France pointing to AppGateFrance-pip. US-endpoint shows Online status.

Name	Status	Monitor Status	Type
US-endpoint	Enabled	Online	Azure endpoint
France-endpoint	Enabled	CheckingEndpoint	Azure endpoint

Resource Visualizer

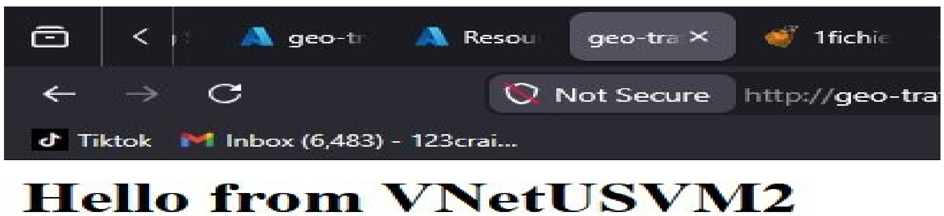
The resource visualizer diagrams below show all resources deployed in each resource group and how they are connected to each other.





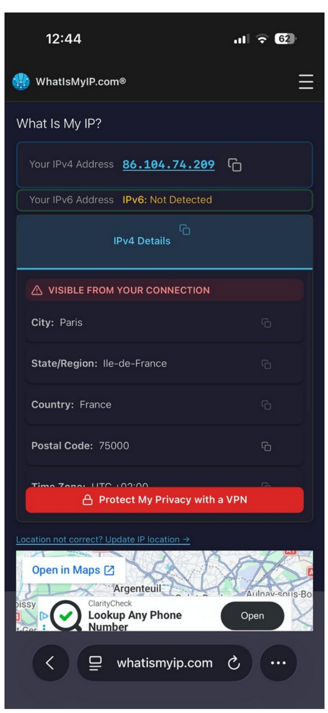
Desktop Browser Test - US Load Balancing

Accessing the Traffic Manager DNS from a US connection routed traffic to AppGateUS. Refreshing the page confirmed load balancing between both US VMs. The screenshots below show VNetUSVM1 and VNetUSVM2 responding on alternate requests.



Mobile Phone Test - France Geographic Routing

Using a VPN set to a French IP address (86.104.74.209, Paris, Ile-de-France), the Traffic Manager DNS was accessed from a mobile browser. The request was correctly routed to AppGateFrance and both VNetFranceVM1 and VNetFranceVM2 responded on separate requests, confirming the geographic filter is working correctly.



Challenges and Solutions

Invalid VNet Address Space

When creating VNetUS, the address space was entered as 10.0.1.0/16 instead of 10.0.0.0/16. Azure rejected this because 10.0.1.0/16 is not a valid CIDR block since the host bits are not zero. Correcting the starting address to 10.0.0.0 resolved the error.

VMs Had No Outbound Internet Access

After deploying VNetUSVM1 and VNetFranceVM1 with no public IPs, running sudo apt update in the Serial Console timed out with connection errors to azure.archive.ubuntu.com. Private VMs have no outbound internet path by default. A NAT Gateway was created for each region and attached to the default subnet, which gave the VMs outbound access for installing Apache.

AppGateUS Public IP Not Selectable in Traffic Manager

When adding the Application Gateway public IP as a Traffic Manager endpoint, it did not appear in the dropdown. Traffic Manager requires the target public IP to have a DNS name label assigned before it can be registered as an Azure endpoint. Adding the DNS name label to both AppGateUS-pip and AppGateFrance-pip through the Public IP Configuration blade resolved the issue.

Portal Link Not Navigating to Public IP Resource

Inside the AppGateUS Frontend IP Configurations panel, clicking the public IP link did not navigate to the Public IP resource page. This appears to be an Azure portal rendering issue. The workaround was to search for Public IP addresses directly in the top search bar and open AppGateUS-pip from there.

What I Learned

This lab extended the Application Gateway setup from the previous assignment into a multi-region architecture. Having two separate VNets, two Application Gateways, and four VMs all unified under a single Traffic Manager DNS entry showed how Azure handles geographic traffic distribution at the DNS level rather than at the network level. Traffic Manager does not proxy or forward packets. It only resolves DNS differently based on the client IP location, so the actual connection goes directly from the client to whichever Application Gateway the DNS resolved to.

The NAT Gateway requirement was an important practical lesson. Private VMs with no public IP have no outbound internet path by default, which blocks package installation. Adding a NAT Gateway to the subnet gives VMs a shared outbound IP for internet access without exposing them individually. This is the correct pattern for secure VM deployments in Azure.

The DNS name label requirement on public IPs for Traffic Manager endpoint registration was encountered again from the previous lab. Standard SKU public IPs must have a DNS name label before Traffic Manager will accept them as Azure endpoints. This is a consistent Azure dependency that applies any time a public IP needs to be referenced by a DNS-based service.

Conclusion

This lab successfully implemented a geographically distributed load balancing architecture using two Azure Application Gateways and Traffic Manager with Geographic routing. US clients accessing the Traffic Manager DNS at geo-traffic-lab.trafficmanager.net are routed to AppGateUS, which load balances between VNetUSVM1 and VNetUSVM2. Clients connecting from France are routed to AppGateFrance, which load balances between VNetFranceVM1 and VNetFranceVM2. Testing from a desktop browser confirmed load balancing on the US side, and testing from a mobile phone with a French VPN IP confirmed the geographic filter correctly routes to the France Application Gateway.